Rift Runner — Game Design Document

Event: Artificial Arcade (24-hour hackathon, Atharva University, Mumbai)

Problem statement: PS01 GlitchVerse (single-player puzzle and adventure)

Team: 3 members, all beginners in game development

Version: 3.0 — pivoted from a turn-based grid to a **2D side-scrolling platformer**; scope locked to 4 levels, built in Godot 4

> *Working title. Alternatives: Parallax Pilot, Source Layer.*

> ***What changed in v3.0.*** *The game is no longer turn-based and grid-based. Nova now **runs and jumps** through side-scrolling levels with real physics (gravity, momentum, continuous collision). The hook is unchanged: edit the hidden **Source Layer** to rewrite reality. Everything in this document that used to say "one tile per turn" now means "free, continuous movement." Level data is no longer an ASCII grid; it is a list of **platforms and entities** with pixel positions.*

---

1. One-line pitch

An astronaut stranded across four planets rewrites reality from a hidden "Source Layer" — running, jumping and spending energy to delete, duplicate and move objects — so she can collect her ship's parts and escape.

2. Story

Astronaut **Nova** crash-lands and her ship breaks apart. The broken parts are scattered across four planets, and every planet is glitched: it exists as two linked dimensions.

- **Dimension A — Reality.** The dangerous world: spikes, pits, walls, moving monsters.
- **Dimension B — Source Layer.** A blue wireframe copy of the same level where objects can be edited. Whatever Nova changes in B changes in A.

Collect every part on a planet, reach the exit portal, and travel to the next planet. After Planet 4, the ship is rebuilt and Nova flies home.

3. Design pillars

1. **Edit reality.** The hook of the game is that Dimension B rewrites Dimension A. Everything else supports this.
2. **Every point is a choice.** One currency, with tight decisions about what to spend on.
3. **Simple to learn, good to feel.** Tight, responsive run-and-jump controls that anyone can pick up in seconds — the challenge is the puzzle, not fighting the controls.
4. **Small but polished.** Four well-tested levels beat ten half-working ones.

4. Core gameplay loop

1. **Run up** to the obstacle in Dimension A.
2. **Enter a portal** to Dimension B.
3. **Reach the B-room objective** to earn points (for example, push a block onto a target pad).
4. **Spend points** on a skill to edit the matching object (delete a spike, duplicate a block into a bridge, move a crate).
5. **Return to A**, where the change has taken effect.

6. **Collect the ship part**, reach the exit portal, and clear the level.

5. Core rules

Rule	Detail
Movement	**Continuous and physics-based.** The player runs left/right with momentum and jumps with gravity (variable jump height). No grid, no turns.
Monsters	Patrol continuously along a set path or between two points; they do **not** wait for the player.
Worlds	Each level is two linked side-scrolling layouts, A and B, with the **same geometry**. Objects have shared IDs, so an edit to the B object changes its A twin.
Portal	Entering a portal zone switches worlds. It is free and unlimited.
Lives	**3 lives per level.** Losing all 3 restarts the level with its starting points. Lives refill at the start of each new level.
Points	One currency called **Energy Points (EP)**. Points can't drop below 0, and a skill you can't afford is locked.
Death	Touching a spike or a monster, or **falling into a pit / off the level**, costs **1 life and 20 EP**. You respawn at the last portal you used (or the level start). Edits you already made stay.
Win	Collect all required ship parts and reach the exit portal.
Lose	Lives reach 0. Show Game Over, with a retry.

6. Skills

Skill	Where used	Cost	Effect
Delete	Dimension B	15 EP	Removes the selected object, and its twin in A disappears (spike, wall, monster).
Duplicate	Dimension B	20 EP	Copies a block into an empty space, and the copy appears in A (a bridge over a pit, a stepping platform).
Move	Dimension B	10 EP	Slides an object to a new spot, and it moves in A too (a crate onto a pressure plate).
Shield	Dimension A	10 EP	Blocks the next hit from a monster or spike (one use).

Targeting (new in v3.0): because there is no grid cursor, a skill acts on the **nearest editable twin within range** of the player. In Dimension B, editable twins glow; walk up to one and press the skill key. If nothing editable is in range, the skill does nothing and the HUD says so.

Stretch skills (only if time remains): Light level (reveals hidden areas), Freeze (stops monsters for 2 seconds).

7. Point economy

Earning

Action	EP
Complete a B-room objective	+25
Complete it within the B-room time limit (bonus)	+10
Pick up an energy cell (A or B)	+5
Clear an obstacle with a skill	+10

Action	EP
Collect a ship part	+30
Finish the level with lives left	+10 per life

Spending and losing

Action	EP
Delete	–15
Duplicate	–20
Move	–10
Shield	–10
Dying	–20 (and –1 life)

Fairness rules

- Each level starts with **30 EP**.
- Every level has at least one route that is winnable with the starting points plus the B objective reward, even after one death.
- Levels 1 to 3 are generous. Only Level 4 is deliberately tight, so the player must choose what to bypass.
- Points never go negative.

Final score

```
Final score = EP remaining at the end of Level 4 + (lives left in each level × 10)
```

Stars: 1 star for finishing, 2 stars for a good score, 3 stars for a high score. Exact thresholds are set after playtesting.

---

8. Levels

Four levels only. Difficulty rises gradually, and each planet adds one new idea. Each level is a **short side-scrolling stage** (about 2–4 screens wide) playable in 1–3 minutes.

Level 1 — Moon (tutorial)

- **New idea:** running, jumping, portal travel and Delete.
- **Obstacles (A):** a pit to cross and one spike blocking a ledge.
- **B room:** push a block onto a target pad (+25 EP), then delete the red block that matches the spike (–15 EP) and duplicate a block to bridge the pit (–20 EP).
- **Parts:** Fuel Cell (1).
- **Starting EP:** 30. **Difficulty:** easy. Text hints guide the player.

Side-view sketch (Dimension A):

```
  P>       c       ^S                *F           (O)        [X]
+-----+          +--------------+             +-----+     +------+
|start|__pit_____|     main     |_____________|ledge|     | exit |
+-----+          +--------------+             +-----+     +------+
```

Legend: `P` spawn, `c` energy cell, `^S` spike, `*F` ship part, `(O)` portal, `[X]` exit, `_pit_` fall-death gap, `+--+` solid platform. Duplicate bridges the pit; Delete removes the spike.

Level 2 — Mars

- **New idea:** Duplicate as the main tool, plus a higher jump gap.
- **Obstacles (A):** a wide pit (broken bridge) and a rock wall blocking the path.
- **B room:** move a block to complete a pattern, then duplicate a block to bridge the pit and delete the wall block.
- **Parts:** Navigation Chip (1).
- **Starting EP:** 30. **Difficulty:** easy to medium.
- **Points check:** Duplicate 20 + Delete 15 = 35, and the B-room reward gives enough.

Level 3 — Europa (ice moon)

- **New idea:** Move, patrolling monsters and Shield.
- **Obstacles (A):** one patrolling monster, a pressure-plate door and a pit.
- **B room:** move a block onto the plate, duplicate for the pit, then either delete the monster or use Shield to run past it.
- **Parts:** Thruster (2, both required).
- **Starting EP:** 30. **Difficulty:** medium.
- **Points check:** Move 10 + Duplicate 20 + (Delete 15 or Shield 10) = 40 to 45. The player must collect energy cells, and a death makes it tight.

Level 4 — Glitch Core (final)

- **New idea:** everything combined, with scarce points.
- **Obstacles (A):** two monsters, spikes, a pit, a locked door and fake "glitch" platforms that drop away (a death) if you stand on them.
- **B rooms:** two Source-Layer objectives to complete in order.
- **Parts:** Core Crystal (1) plus the final launch gate.
- **Starting EP:** 30. **Difficulty:** hard.
- **Points check:** using every skill would cost more than the player can earn, so the player must decide which obstacles to remove and which to jump past.

Difficulty curve

Level	Planet	Skills in play	Hazards	Difficulty
1	Moon	Delete, Duplicate	Spike, pit	Easy
2	Mars	Delete, Duplicate	Wide pit, wall	Easy to medium
3	Europa	Delete, Duplicate, Move, Shield	Monster, plate, pit	Medium
4	Glitch Core	All	Two monsters, spikes, pit, door, fake platforms	Hard

---

9. Objects and obstacles

Object	In A	In B	Notes
Platform	Solid ground to stand/run on	Same solid, drawn as wireframe	Shared geometry; the level's floor and walls

Object	In A	In B	Notes
Wall	Blocks the path	Editable block	Can be deleted or moved if marked editable
Spike	Kills on contact	Red editable block	Delete removes it
Pit / broken bridge	Fatal to fall into	Empty gap (walkable wireframe floor in B)	Duplicate a block into it to make a bridge in A
Crate	Pushable / movable	Editable block	Move places it on plates
Pressure plate	Opens a door / drops a barrier when a crate is on it	Target pad	Needs a crate (via Move) or the player
Monster	Patrols a fixed path, kills on contact	Editable block	Can be deleted, shielded against, or jumped over
Energy cell	+5 EP	+5 EP	Optional pickup
Ship part	Required pickup	Not present	Counts toward the level goal
Portal	Switches worlds	Switches worlds	Free to use; a zone you enter
Exit	Ends the level	Not present	Only works with all parts collected

10. Screens and interface

1. **Title** — game name, Play, Credits.
2. **Level select** — 4 planets, locked until the previous one is cleared.
3. **Gameplay** — side-scrolling stage with a camera that follows the player, portal toggle, glitch effect on world change.
4. **HUD** — EP, lives (3 icons), parts collected, skill buttons with costs, current planet name.
5. **Level clear** — points breakdown and Next.
6. **Game over** — Retry, Level select.
7. **Ending** — the ship launches, with the final score and stars.

Feedback: floating `+10` and `−20` numbers, a screen shake on death, a blue wireframe tint plus scanlines in Dimension B, and a short glitch effect when travelling through a portal.

11. Art and audio direction

- Clean 2D side-scrolling look, space theme, with one astronaut (idle, run, jump). Reference images provided by the team drive the final style (vector/flat silhouette direction).
- **Parallax backgrounds** (stars, a planet) that scroll slower than the foreground for depth.
- One shared set of platform/hazard art, **re-skinned by color** per planet: grey (Moon), red (Mars), icy blue (Europa), purple-black (Glitch Core).
- Dimension B uses the same shapes with a blue wireframe look, so no new art is needed.
- Sound: footstep, jump, portal, skill use, point gain, death, level clear, and one looping background track.

---

12. Technical approach (proposed)

- **Engine:** Godot 4 with GDScript (free, open source, MIT licence, no fees). 2D side-scroller using Godot's physics. The whole team uses the same Godot 4.x version.

- **Player:** a `CharacterBody2D` moved in `_physics_process(delta)` — gravity, horizontal acceleration/friction, and a jump with coyote time and a short jump buffer for good feel. Collision against platforms uses `move_and_slide()` (or a simple AABB resolve), not grid math.
- **Tools (all free):** Godot, Antigravity (free Individual plan) as the code editor and LLM agent, GitHub for the repo, and free or CC0 art and sound (Kenney, OpenGameArt, Piskel, Krita, jsfxr, Audacity).
- **Separation:** game logic (state, rules, physics, points, lives) is separate from rendering (scenes, sprites, UI). The logic lives in plain GDScript classes behind one `Game` autoload. Scenes only read state, listen to its signals, and feed input such as `set_move` and `jump`.
- **Levels as data:** each level is a JSON file describing the level size, spawn point, a list of **platforms** (solid rects) and a list of **entities** (spikes, pits, monsters, crates, plates, cells, parts, editable twins, portals, exit), with shared twin IDs, starting points, parts, and the B-room objective. Loaded with Godot's built-in JSON parser.
- **Suggested folders (inside the Godot project):** `src/logic/`, `src/levels/`, `src/scenes/`, `src/ui/`, `assets/`, `docs/`. Each folder has one owner, and only one person edits `project.godot`.
- **Scenes:** one small `.tscn` per screen, with most nodes created in script, so scene files stay tiny and merge cleanly in Git.
- **Export:** a Windows build for the demo, plus a Web (HTML5) build as a backup, using Godot's free export templates.
- **Game state holds:** current level, world (A or B), player position **and velocity**, whether the player is on the ground, points, lives, parts collected, entity list, and the level timer.
- **Tuning table** is kept in one config file (`src/logic/config.gd`) — costs, rewards, **and movement values** (gravity, run speed, acceleration, friction, jump velocity, coyote time) — so balance and feel changes never touch the rules.

13. Scope lock

In scope (must have): 4 side-scrolling levels, 2 linked dimensions, continuous run/jump movement, point system, 3 lives per level, Delete, Duplicate, Move, Shield, all 7 screens, basic sound.

Only if time remains: Light level and Freeze skills, extra animations, moving platforms, a score leaderboard, level timers.

Out of scope: turn-based or grid movement, more than 4 levels, real-time combat beyond contact damage, multiplayer, 3D.

14. Testing checklist

- ■ Each level can be completed with the starting points plus rewards, and with one death
- ■ Points never go below 0, and locked skills show as locked
- ■ Losing all 3 lives restarts the level correctly
- ■ An edit in B always shows in A, and the player can't get stuck
- ■ The player can't clip through platforms, and falling into a pit always triggers a death
- ■ Exit only works with all parts collected
- ■ All screens appear in the right order, with no crash

15. Glossary

Term	Meaning
Dimension A / Reality	The hazard world the player runs and jumps through
Dimension B / Source Layer	The editable wireframe world

Term	Meaning
EP	Energy Points, the single currency
Skill	A power-up that costs EP (Delete, Duplicate, Move, Shield)
Part	A ship component the player must collect
Platform	A solid rectangle the player can stand and run on
Coyote time	A short grace window after leaving a ledge during which a jump still works